Formation of Supermassive Black Holes

Krupa Pothiwala & Liz Hayes

Florida Institute of Technology

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Outline

Background Information

Formation of the First Supermassive Black Holes

Formation of SMBHs by Direct Collapse in Pregalactic Halos

SMBH formation via collisions in BH clusters

References

Background Information

- What are supermassive black holes?
  - Black holes with masses above 100,000 $M_\odot$ Wikipedia contributors (2026)
  - Some reach several billion solar masses
- Where are they found?
  1. At the centers of most galaxies
  2. Occasionally wandering through intergalactic space
- How are they formed?
  - This is the question we are exploring

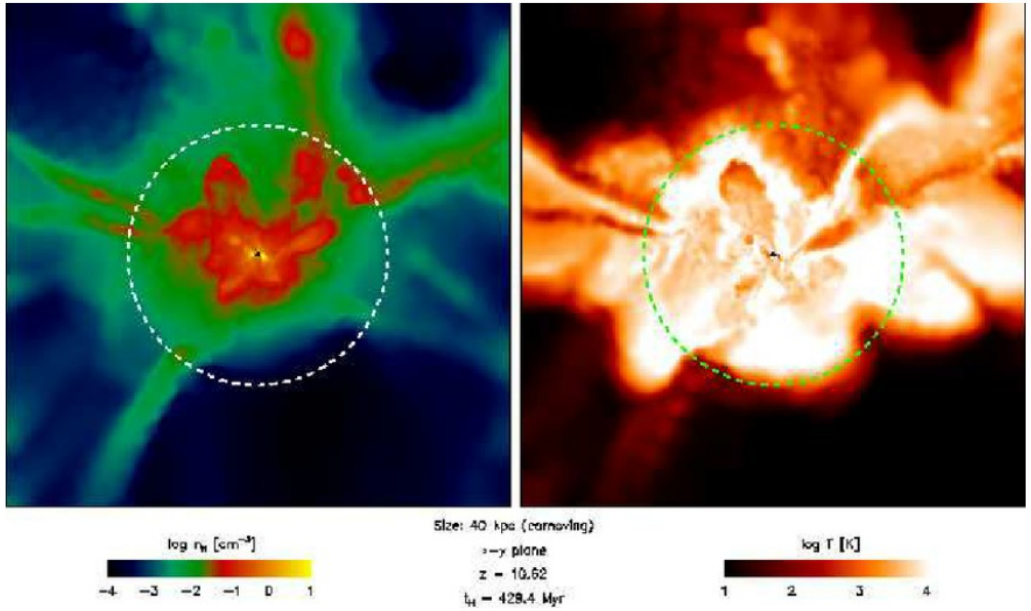
Formation of the First Supermassive Black Holes

Image: Archyde – SMBH: Echoes of the Universe's First stars?

Why Direct Collapse?

- Quasars are observed at redshift $z > 6$, less than 1 billion years after the Big Bang
- These quasars already host black holes with masses $\sim 10^9\, M_\odot$
- Stellar-mass seeds (10–100 $M_\odot$) may not grow quickly enough
- Bromm and Loeb (2003) explores whether gas clouds could collapse directly into massive seeds

Key Conditions for Direct Collapse



The properties of the primordial gas collapsing into an atomic cooling dark matter halo at z ≃ 10. Shown are the hydrogen number density (left panel) and temperature (right panel), the dashed lines denoting the virial radius r vir at a distance of ≃ 1 kpc. Note that most of the gas is accreted directly from the IGM and shock-heated to the virial temperature of T vir ≃ 10 4 K, although cold accretion also becomes

- Gas must be nearly metal-free
- Strong ultraviolet radiation destroys molecular hydrogen (H_2)
- H_2 normally allows primordial gas to cool
- Without H_2, gas stays hot ($\sim 10^4$ K)
- Hot gas prevents fragmentation into many stars
- Gas collapses more smoothly toward the halo center

Simulation Setup

- Simulates collapse of a $\sim 10^8\, M_\odot$ halo at $z \sim 10$
- Uses Smoothed Particle Hydrodynamics (SPH)
- Tracks gas motion and cooling
- Chemical network follows hydrogen and helium species
- Compares cases with and without H_2 cooling

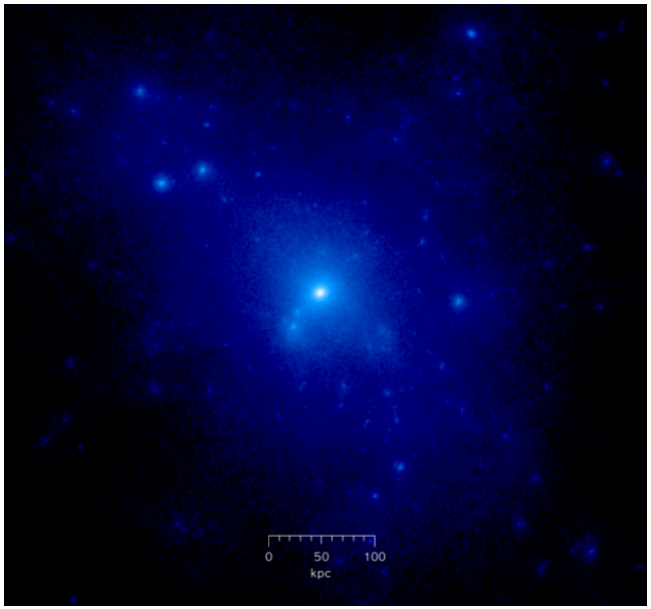


Figure: Simulated dark matter halo from a cosmological N-body simulation. Taken from Wikipedia.

Main Result

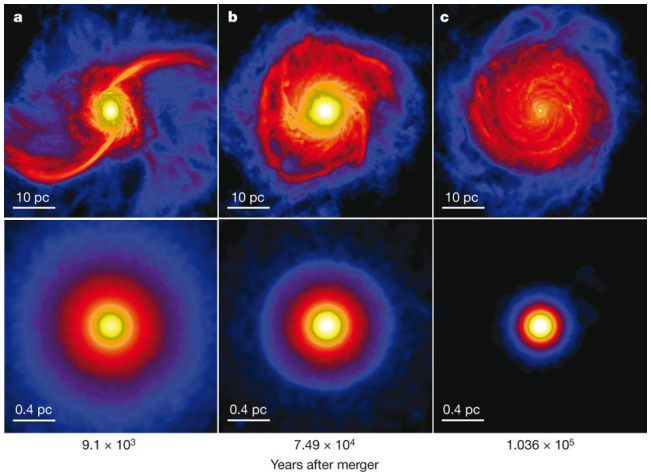


Figure: Time evolution of the nuclear gas disk from its formation until the onset of central collapse. Taken from Nature article

- Without H_2 cooling, gas remains near $\sim 10^4$ K
- Fragmentation into normal stars is suppressed
- Gas collapses toward the halo center
- A central clump forms with mass $\sim 10^6\ M_\odot$
- More than 10% of halo gas can collect in the nucleus

Takeaways from Source 1

- Direct collapse can form massive black hole seeds quickly
- Typical seed masses are $\sim 10^5$–$10^6\ M_\odot$
- This helps explain quasars observed at $z > 6$
- Requires rare conditions:
  - metal-free gas
  - strong UV radiation
  - massive halos

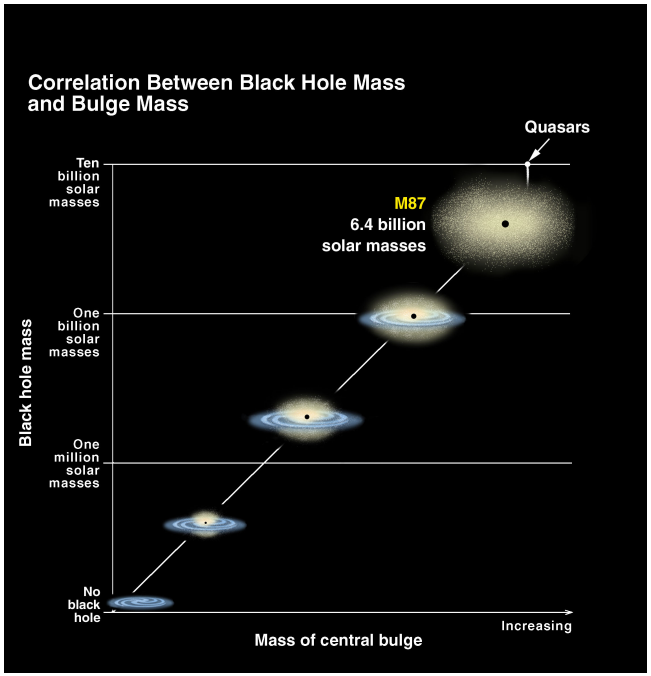
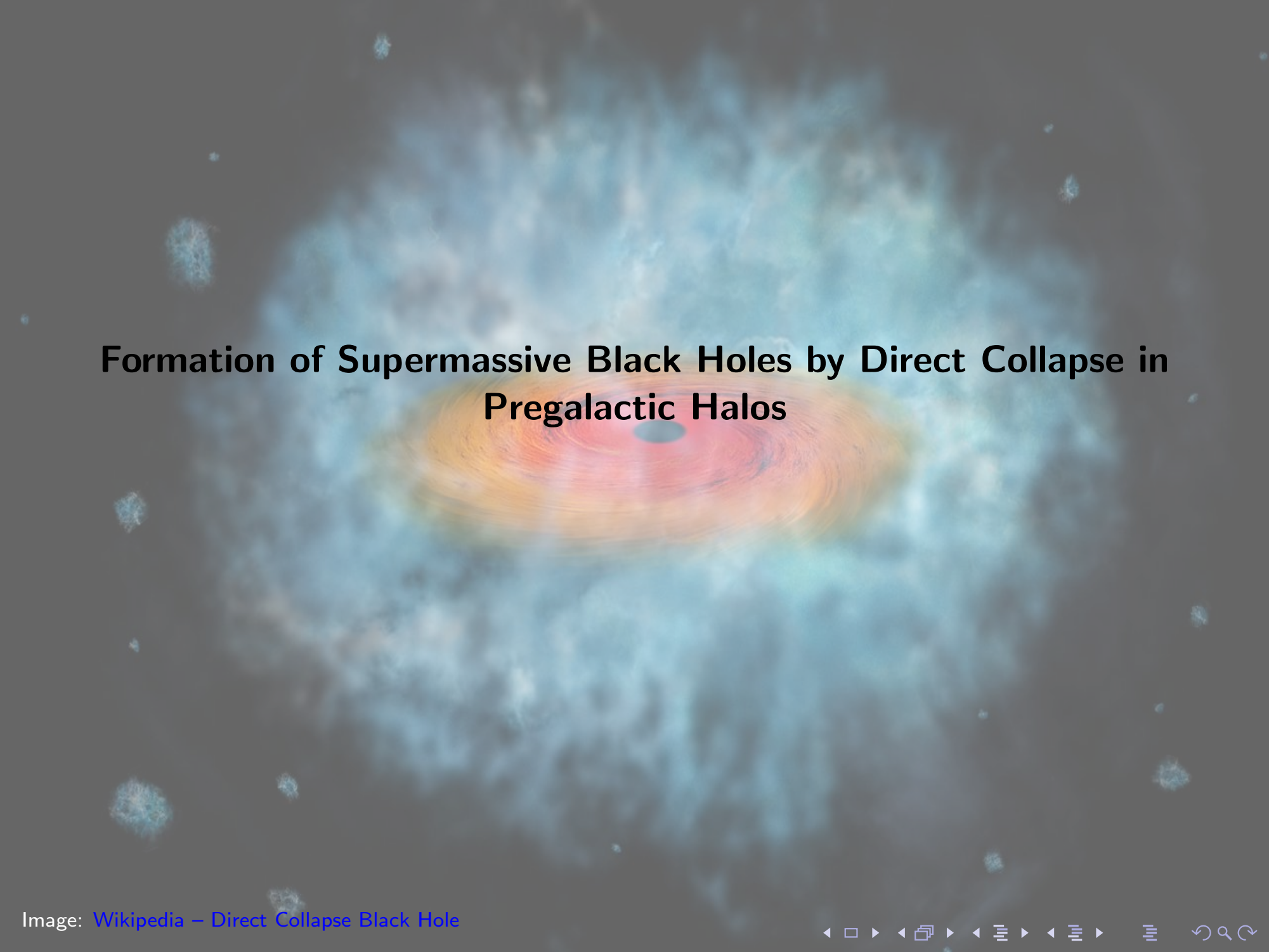


Figure: The illustration shows the relationship between the mass of a galaxy's central black hole and the mass of its central bulge. Taken from The University of Texas McDonald Observatory



Formation of Supermassive Black Holes by Direct Collapse in Pregalactic Halos

Image: Wikipedia – Direct Collapse Black Hole

Formation directly in the nuclei of protogalaxies

- How is it possible to generate a SMBH at $z \sim 6$?
- Begelman et al. (2006) proposes that SMBHs form directly from gas in early halos
- Halos with virial temperature $\sim 10^4$ K allow atomic hydrogen cooling
- Gas begins collapsing toward the halo center
- Angular momentum must be removed for collapse to continue

Bars Within Bars

- Self-gravitation gas clouds become bar-unstable.
- Bars transfer angular momentum outward through gravitational and hydrodynamical torques causing the radius to shrink.
- The bar instability causes the gas/mass to fall inward
- For this to hold there needs to metal free has in order to avoid gas fragmentation

Quasi-Star

- Gas accumulates rapidly in the halo center
- A massive object called a quasi-star forms
- The envelope is supported by radiation pressure
- Pressure decreases and causes core collapse
- Rapid accretion allows fast black hole growth

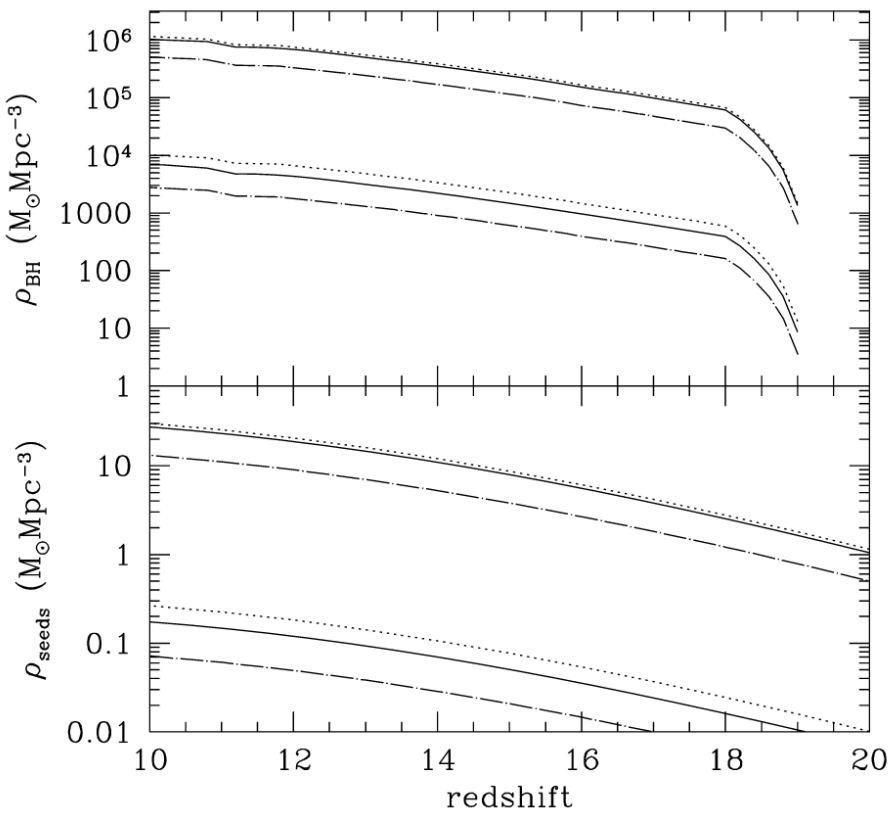


Figure: Comoving density of BH seeds (bottom) growing to $10^6\ M_\odot$ (top), from $z = 20$ to $z = 10$ (Begelman et al., 2006).

Supermassive Black Hole Formation via Collisions in Black Hole Clusters

Image: Nasa Black Hole Merger Simulation

Dense Cluster Formation

- The universe is known to host Nuclear Star Clusters where SMBHs co-exist Gaete et al. (2024)
- Fragmentation causes stars to be packed very close together this increases stellar density
- Due to high stellar density the cluster can undergo runaway core collapse, causing an intermediate-mass black hole ($\sim 10^{2-4}$ $M_\odot$)
- The BH could be surrounded by inflow of gas increasing the gravitational potential of the cluster.
- As well as dynamical friction will increase the gravitational potential

Thought Experiment

- Consider a cluster made of stellar-mass black holes and stars
- The BH will start forming Binaries. These binaries are going to interact with other black holes. What do you think will happen?
- This causes the cluster to stabilize and not collapse in on itself
- This is how velocity dispersion holds off gravitational collapse

Eventually the binary gets close enough and starts radiating gravitational waves and merges into a single black hole increasing the average black hole mass in the cluster

Forming a SMBH

- Since the binary merged the gravity overpower the velocity dispersion
- During this phase: density rises dramatically, encounter rates increase, and mergers become much more frequent.
- A runaway growth process begins where one black hole grows rapidly through repeated mergers. This produces a massive central black hole.

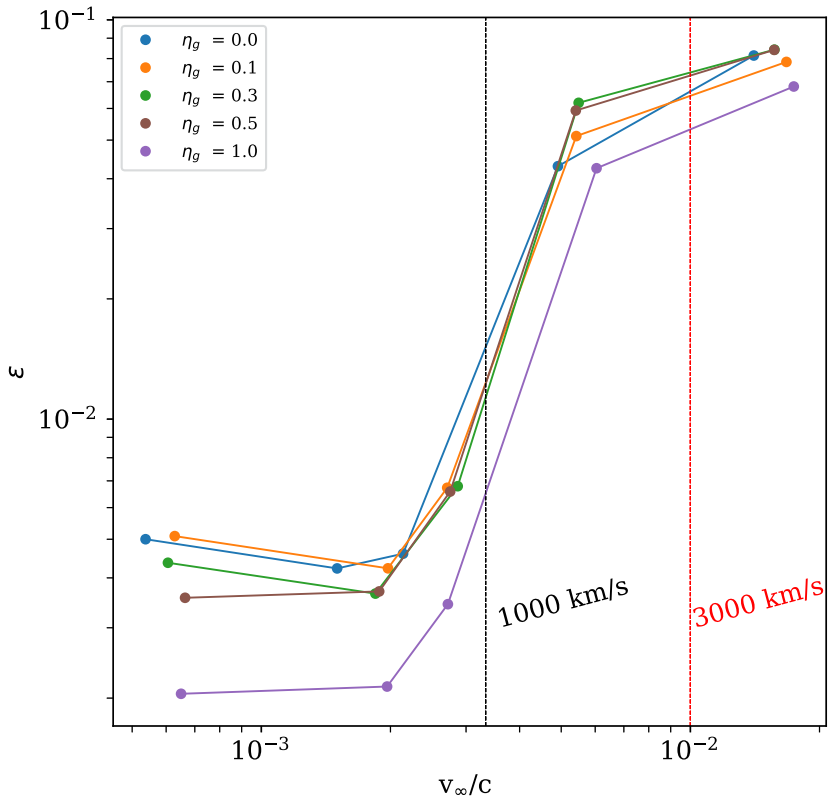


Figure: Formation efficiency of the most massive BH as a function of cluster velocity relative to the speed of light (Gaete et al., 2024).

References

Begelman, M. C., Volonteri, M., and Rees, M. J. (2006). Formation of supermassive black holes by direct collapse in pregalactic halos. *Monthly Notices of the Royal Astronomical Society*.

Bromm, V. and Loeb, A. (2003). Formation of the first supermassive black holes. *The Astrophysical Journal*.

Gaete, B., Schleicher, D. R., Lupi, A., Reinoso, B., Fellhauer, M., and Vergara, M. C. (2024). Supermassive black hole formation via collisions in black hole clusters. *arXiv preprint*.

Wikipedia contributors (2026). Black hole. https://en.wikipedia.org/wiki/Black_hole. Accessed: March 12, 2026.